

Victor Gaya

AI Research Engineer | Multimodal AI & Representation Learning

Profile

AI Research Engineer (PhD, IP Paris) specializing in **Computer Vision**, **Multimodal Foundation Models**, and **Spatial Reasoning**. Proven record in self-supervised learning, cross-modal optical/radar fusion, and physics-grounded AI agents. Combines deep geometric rigor with rapid PyTorch execution.

Contact

🌐 goldentrex.github.io
✉️ vgaya@nus.edu.sg
☎️ +33 6 45 92 39 71
🌐 LinkedIn: Victor Gaya
📄 ResearchGate: Victor Gaya
🆔 ORCID: 0009-0004-2700-7760

Languages

🇫🇷 French – Native Language
🇬🇧 English – Fluent (TOEIC C1, 2022)

Technical Skills

💡 VLMs, Diffusion Models, GenAI
👁️ Spatial Reasoning, 3D Vision, SSL
🔗 Physics-Grounded AI, Complex Math
📄 Python, PyTorch, HuggingFace, CUDA
📦 Distributed Training, W&B, Docker
🚀 Agentic AI, Tool-Use, APIs, Slurm

Involvements

★ **Agentic AI Systems:** Built autonomous LLM/VLM multi-agent workflows with tool-use in hackathons
🚀 **Drone Club:** Founded aerial robotics & computer vision association at CEA
👤 **Leadership:** President of student associations (managed budgets, 50+ person team coordination)

Other Interests

🎵 Music production 🏸 Tennis
⚡ Mountaineering 🧗 Climbing

WORK EXPERIENCE

Jan. 2026 - Present



Research Fellow – Multimodal AI
NUS (National Univ. of Singapore)

📍 Singapore

ECE Department

- **Multimodal Foundation Models:** Research on **VLMs** and **diffusion models** for cross-modal registration and image translation between optical and SAR imagery.
- **Physics-Grounded AI:** Developing physics-informed models and complex-valued representations (polarimetry, phase distributions) for high-dimensional visual reasoning.
- **Research Co-Supervision:** Mentoring graduate students on experimental design, InSAR 3D workflows, distributed **PyTorch**, and **W&B** tracking.

Dec. 2022 - Dec. 2025



Research Engineer (PhD)
CEA & Télécom Paris (IP Paris)

📍 Paris, France

AI Research Group

- **Multimodal Self-Supervised Learning:** Designed self-supervised deep learning architectures for optical and SAR image fusion and despeckling without paired ground truth.
- **InSAR Phase Denoising & 3D:** Formulated neural models for InSAR phase denoising and 3D surface reconstruction grounded in statistical noise distributions.
- **Dissemination:** Published and presented peer-reviewed discoveries at premier international conferences and journals (**IGARSS 2024**, **JSTARS 2026**).

Mar. 2022 - Sept. 2022



Machine Learning Engineer Intern 📍 Toulouse, France
French Space Agency (CNES) *Autonomous Systems & HITL*

- **Human-in-the-Loop Anomaly Detection:** Engineered real-time anomaly detection pipelines, integrating expert user feedback loops on operational satellite telemetry.
- **Operational Missions:** Deployed and validated ML models on operational space missions meeting stringent reliability and latency SLAs.

May 2021 - Aug. 2021



Computer Vision Engineer Intern 📍 Marseille, France
KORNER (KLANIK) *Edge AI Systems*

- Trained and deployed a real-time convolutional neural network on streaming video for automated object detection in field-deployed devices across France.

EDUCATION

2022 - 2025



PhD in Computer Vision & AI
Institut Polytechnique de Paris

📍 Paris, France

Télécom Paris

Focus: Multimodal representation learning, self-supervised pretraining, and 3D geometric vision for complex surface reconstruction.

2016 - 2022



MSc & Engineering Degree in AI
JUNIA (ISEN Lille)

📍 Lille, France

Applied Math & Computer Science

Specialization: Artificial Intelligence & Data Science. Rigorous foundations in statistical learning, image processing, and deep neural networks.